# Topic 10: Prompt Chaining and Multi-Step Tasks

## 10.1 What is Prompt Chaining?

- Prompt chaining: breaking a complex task into sequential steps where output of one prompt becomes input to the next
- Each step is simpler, more focused, and easier to validate than one giant prompt
- Enables quality checkpoints between steps -- catch errors before they propagate
- Used in: document processing pipelines, multi-stage analysis, content generation workflows
- Think of it as a pipeline: each prompt is one transformation in the data flow

## 10.2 Designing a Prompt Chain

- Step 1: Map out the full task end-to-end before writing any prompts
- Step 2: Identify natural breakpoints where output can be cleanly passed to the next step
- Step 3: Write and test each prompt independently before connecting them
- Step 4: Define the data contract between steps -- what format does each step output/expect?
- Step 5: Add validation between steps -- check output quality before passing it forward

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Example: Automated blog post pipeline

Step 1 -- Research prompt:
  Input: topic keyword
  Output: 5 key facts and statistics (JSON array)

Step 2 -- Outline prompt:
  Input: facts from Step 1
  Output: blog post outline (5 sections, 3 points each)

Step 3 -- Draft prompt:
  Input: outline from Step 2
  Output: full draft (~800 words)

Step 4 -- Edit prompt:
  Input: draft from Step 3
  Output: polished version with SEO improvements
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## 10.3 Parallel Prompting

- Some sub-tasks are independent -- run them in parallel to reduce latency
- Example: analyse sentiment, extract keywords, AND identify entities from the same text simultaneously

- Merge parallel outputs in a final consolidation prompt
- Reduces total wall-clock time when using async API calls

### 10.4 Self-Checking Chains

- Add a verification prompt after key steps: 'Review the output above. Is it accurate and complete? If not, correct it.'
- Reflection prompting: ask the model to critique its own answer, then produce an improved version
- Useful for reducing hallucinations in factual tasks

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--- Step 1: Generate answer ---
Answer the question: [question]

--- Step 2: Self-check (new prompt call) ---
You previously answered a question. Here is your answer:
[insert Step 1 output]

Review your answer for:
1. Factual accuracy
2. Completeness
3. Any logical errors

If the answer is correct and complete, reply 'VERIFIED'.
Otherwise, provide a corrected answer.
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